

Credit Scoring Model using Machine Learning

CodeAlpha Machine Learning Internship - Task 1

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Project: Credit Scoring Model

Domain: Machine Learning

Abstract

This project develops a machine learning based credit scoring model to predict an individual's creditworthiness using financial history. The model classifies customers into Good, Standard, and Poor credit score categories. The project includes data preprocessing, feature engineering, classification model training, performance evaluation, visualization, and model saving.

Objective

The main objective is to predict creditworthiness using customer financial data such as income, debts, credit utilization ratio, delayed payments, credit history age, and monthly balance.

Dataset Description

The dataset contains 2000 records and 15 columns. The target variable is Credit_Score, which contains three classes: Good, Standard, and Poor.

Feature	Description
Age	Customer age
Annual_Income	Annual income of the customer
Monthly_Inhand_Salary	Monthly salary after deductions
Num_Bank_Accounts	Number of bank accounts
Num_Credit_Card	Number of credit cards
Interest_Rate	Loan or credit interest rate
Num_of_Loan	Number of active loans
Delay_from_due_date	Delay in payment from due date
Outstanding_Debt	Total outstanding debt
Credit_Utilization_Ratio	Credit usage percentage
Credit_History_Age	Age of credit history
Total_EMI_per_month	Monthly EMI amount
Monthly_Balance	Monthly remaining balance
Credit_Score	Target class

Methodology

- Load the dataset from CSV file.
- Remove duplicate rows and handle missing values.

- Encode categorical values using LabelEncoder.
- Split data into training and testing sets.
- Apply feature scaling for Logistic Regression.
- Train Logistic Regression, Decision Tree, and Random Forest models.
- Evaluate models using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
- Save the best model using Joblib.

Algorithms Used

Three classification algorithms were used: Logistic Regression, Decision Tree Classifier, and Random Forest Classifier. The best model was selected based on accuracy.

Results

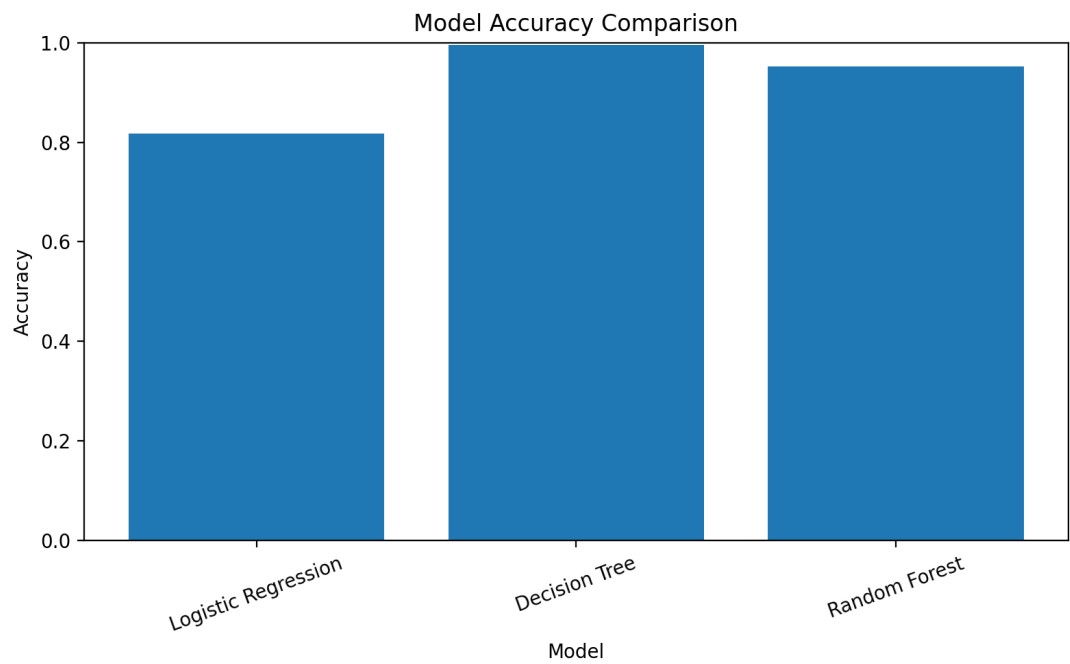
Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.8175	0.8156	0.8175	0.8164
Decision Tree	0.9950	0.9950	0.9950	0.9950
Random Forest	0.9525	0.9548	0.9525	0.9500

Best Model: Decision Tree

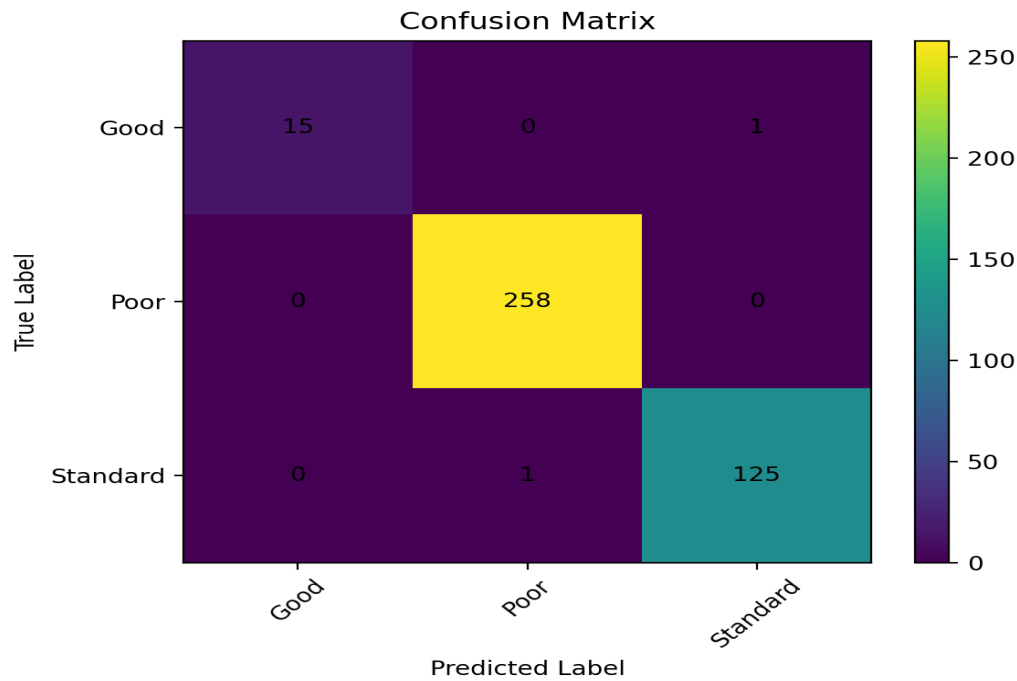
ROC-AUC Score: 0.9865

Project Visualizations

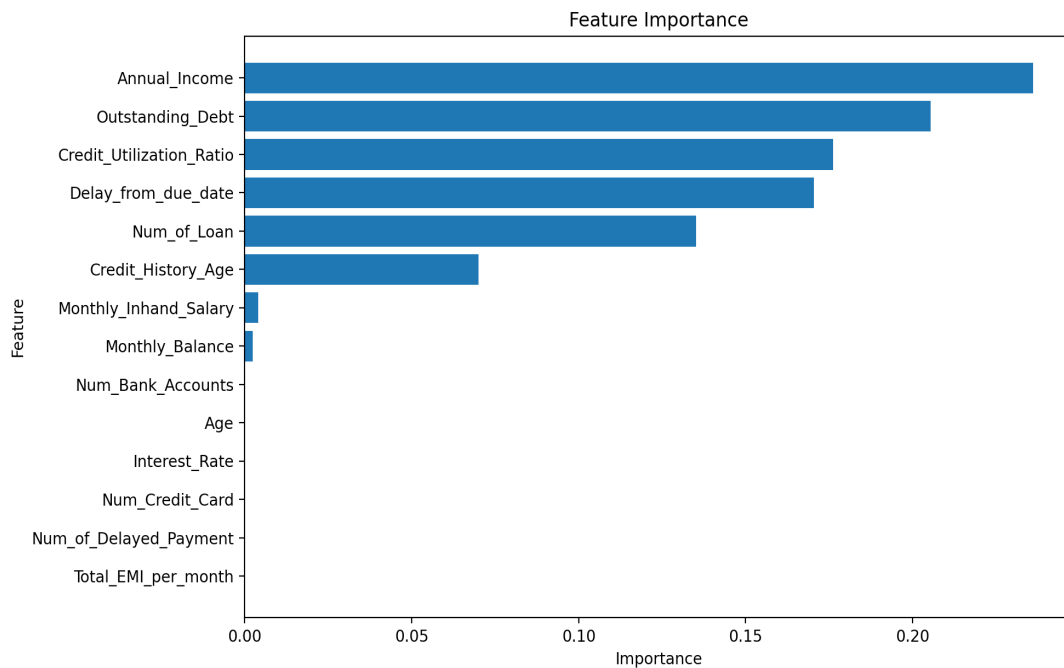
Model Accuracy Comparison



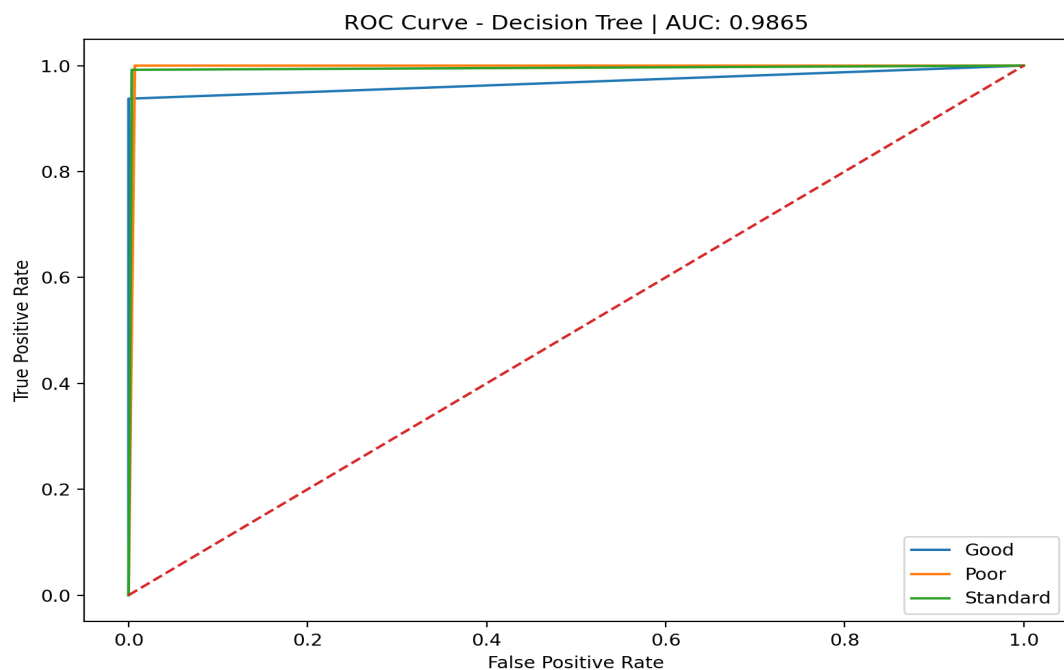
Confusion Matrix



Feature Importance



ROC Curve



Conclusion

The Credit Scoring Model successfully predicts customer creditworthiness based on financial attributes. Random Forest generally performs well because it handles non-linear relationships and feature interactions effectively. The final trained model, scaler, and label encoders were saved for future prediction and deployment.

Future Scope

- Use a larger real-world banking dataset.
- Add hyperparameter tuning using GridSearchCV.
- Deploy the model using Streamlit or Flask.

- Add XGBoost or LightGBM for improved performance.
- Build a user interface for credit score prediction.